

Introduction to Computer Vision

Universidad Católica del Norte

Carlos H. Menacho-Guerra

Departamento de Ingeniería de Sistemas y Computación

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Welcome!

Before we dive in, let's go around the room:

- Name and program/year
- Research interest/application
- Why did you take this signature?
- What do you hope to get from this course?

(Be yourself!)

Welcome!

Let's define some rules:

- Questions are welcome at any time — interrupt me
- Laptops open for notes; slides posted after class
- Running late? Come in anyway — just take the nearest seat quietly; There is a 10-minute grace period for attendance.
- Any plagiarized work will receive a 0.
- AI is permitted, but students must be able to explain their solutions.

We will cover the following topics:

- Introduction and computer vision pipeline
- Sensors and Image formation
- Image processing
- Edges and morphology
- Feature extraction
- Coordinate systems and transformations
- Pinhole model
- Camera calibration
- Basic measure and 3D reconstruction
- Simulation software

We will use the following tools:

- Conda or miniconda – We will use the same environment
- Python programming language
- Numpy
- Rerun-SDK – Visualization and Data Base

Quick Diagnostic Test

Before we start, please scan the QR code and complete the short diagnostic questionnaire
(takes about 10 minutes — it is not graded)



Session Agenda (3 hours)

#	Block
1	Welcome & logistics
2	What is computer vision?
3	Marr's levels of description
4	Applications: industrial & consumer
5	A brief history of computer vision
6	Book & course roadmap
7	Notation conventions
8	Sample syllabus
9	Wrap-up, discussion & references

Content follows Szeliski (2022), Ch. 1 [1], supplemented with primary sources cited throughout and collected on the final slides.

By the end of this session you should be able to:

- Explain why computer vision is an **inverse problem** — and why that makes it hard
- Describe the main approaches used in the field: **scientific, statistical, engineering, and data-driven**
- Explain **Marr's three levels of description**
- Place a given technique on the **50-year timeline** of computer vision
- Understand how the textbook's chapters connect to each other
- Recognize the book's **notation** well enough to follow later chapters

Outline

- 1 What Is Computer Vision?
- 2 Marr's Levels of Description
- 3 Applications of Computer Vision
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Human Vision vs. Computer Vision

Human perception is effortless:

- Instant 3D structure from a static image (shape, translucency, shading of a flower petal).
- Automatic figure/ground segmentation.
- Immediate recognition of people, objects, emotions in a cluttered scene.

Yet a rigorous computational account of *how* biological vision does this remains open [2, 3, 4, 5, 6].

Computer vision is hard:

- Decades of progress, yet even state-of-the-art systems fail at tasks a two-year-old solves trivially.
- Reliable 3D reconstruction from thousands of photos is now possible [10]; general scene understanding is not.

Discussion prompt

Why does the “AI effect” keep undervaluing perception relative to symbolic reasoning? What does this say about how we benchmark progress?

Vision as an Inverse Problem

Forward model (the domain of physics & computer graphics):

scene geometry, material, lighting $\longrightarrow$ image

Inverse problem (the domain of computer vision):

image(s) $\longrightarrow$ scene geometry, material, lighting

- Inverse problems are typically **ill-posed**: insufficient information to uniquely determine the solution (Hadamard sense – existence, uniqueness, stability can all fail).
- We must inject additional knowledge: physics-based priors, probabilistic priors, or priors learned from data.
- Graphics is comparatively mature (rendering a still scene or an extinct creature can look photoreal); the inverse direction is not.

Why this framing matters

Every technique in this course is best understood as: *what forward model does it assume, and what does it use to regularize the inversion?*

Why Is Vision So Difficult?

- **Loss of depth:** a 2D image is a projection of a 3D scene – many different 3D scenes can produce the exact same image.
- **Nuisance variables:** lighting, viewpoint, occlusion, clutter, and sensor noise all make the same object look different.
- **A huge, complex world to model:** the visual world is far richer and harder to model than, e.g., the sounds of speech.
- **A historical surprise:** early AI researchers assumed vision would be the "easy" part, solved quickly before tackling reasoning and planning (Minsky's 1966 "summer vision project" [7]) – they were wrong, and vision turned out to be one of the hardest problems in AI.

Key idea

Because many 3D scenes can explain the same 2D image, computer vision problems are *ill-posed*: we need extra assumptions (priors) to pick the most likely explanation.

Four Methodological Pillars

Szeliski organizes the entire field around four complementary approaches [1]:

- ❶ **Scientific** – build a detailed forward model of image formation and invert it, simplifying where needed for tractability.
- ❷ **Statistical** – treat scene and noise as random variables; use Bayesian inference to combine priors with a noisy likelihood.
- ❸ **Engineering** – favor simple, well-tested techniques with known failure modes and predictable run-time cost.
- ❹ **Data-driven** – collect representative labeled data to learn or validate model parameters directly.

These are not competitors: modern pipelines routinely combine all four (e.g., a physically-based camera model + a learned prior + a robust estimator, validated on benchmark datasets).

Engineering approach – a recommended validation protocol:

- ① Test on **clean synthetic data** with known ground truth.
- ② Add **controlled noise**; measure graceful degradation.
- ③ Test on **diverse real-world data** (e.g., web photos).

Only step 3 tells you whether an algorithm handles real complexity.

Data-driven approach – representative, sufficiently large, unbiased training data is now often the binding constraint on performance – more so than model architecture.

Discussion prompt

For your own research problem: what would “synthetic $\rightarrow$ noisy $\rightarrow$ real” validation look like? Where does distribution shift most likely break your model?

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David Marr's Three Levels (1982)

Marr's *Vision* [2] remains the field's most influential epistemological framework. Three levels, loosely paraphrased:

- 1 **Computational theory:** What is the goal of the computation, and what constraints (physical, statistical) bear on it?
- 2 **Representation & algorithm:** How are inputs, outputs, and intermediate quantities represented, and by which algorithm are they computed?
- 3 **Hardware implementation:** How are representation and algorithm realized physically – biological neurons, or silicon (GPUs, many-core, neuromorphic chips)?

Why it still matters

A careful *computational-theory* analysis (task, constraints, priors) married to an efficient, robust *algorithm* is exactly the scientific+statistical+engineering synthesis from the previous section.

Applying Marr's Levels: A Worked Example

Task: find the edges in an image (where objects begin and end).

- **Computational theory** — *what* and *why*:
an edge is where brightness changes sharply, so look for places where the image gradient is large.
- **Representation & algorithm** — *how*:
store the image as a grid of pixels; slide a small filter (e.g. Sobel) over it to measure the change, then keep the strong responses.
- **Hardware implementation** — *what runs it*:
the same filtering runs on a CPU, on a GPU, or in the neurons of the visual cortex.

The point

The three levels are **independent**. Two people can agree that edges are large gradients, use different filters to find them, and run those on completely different hardware.

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- **Optical character recognition (OCR):** postal code reading, automatic number-plate recognition.
- **Machine inspection:** stereo vision with specialized illumination for tolerance checks on aircraft wings / auto body parts; X-ray inspection of steel castings.
- **Retail:** object recognition for automated checkout and fully autonomous stores.
- **Warehouse logistics:** autonomous package delivery, pallet-carrying robots, robotic parts picking.

- **Medical imaging:** registering pre-/intra-operative imagery; longitudinal studies of brain morphology.
- **Self-driving vehicles:** point-to-point autonomous driving and autonomous flight.
- **3D model building (photogrammetry):** fully automated 3D reconstruction from aerial/drone imagery.

These domains share a pattern: *narrow, well-controlled imagery + strong domain priors + high reliability requirements.*

- **Match move:** merging CGI with live-action footage by tracking feature points to recover 3D camera motion (e.g. *Jurassic Park*); requires precise matting.
- **Motion capture (mocap):** retro-reflective markers viewed from multiple cameras, or fully markerless vision-based capture.
- **Surveillance:** intruder detection, highway traffic analysis, drowning detection.
- **Biometrics:** fingerprint recognition, access authentication, forensic applications.

- **Stitching:** seamless panoramas from overlapping photos [31] (Ch. 8).
- **Exposure bracketing / HDR:** merging multiple exposures into one well-exposed image (Ch. 10).
- **Morphing:** seamless transitions between two photographs (Ch. 3).

- **3D modeling from snapshots:** converting personal photos into a 3D object/person model (Ch. 13).
- **Video match move & stabilization:** inserting 2D/3D content into video, or removing camera shake (Ch. 9, Ch. 11).
- **Photo-based walkthroughs:** navigating photo collections by flying between views in 3D (Ch. 14).

Consumer Applications III

- **Face detection:** improved autofocus, relevant image search (Ch. 6).
- **Visual authentication:** automatic user login from a webcam (Ch. 6).

Why lead with applications?

Applications turn open-ended inverse problems into well-constrained ones: they tell you which unknowns matter, what data is available, and what error is tolerable. Work *backward* from the problem to the technique, not forward from a favorite technique.

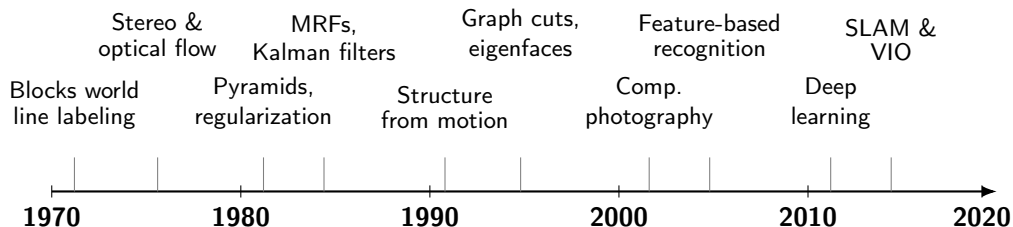
Discussion prompt

Pick an application on this slide (or your own). What real-world constraint would you exploit as a prior that a generic algorithm could not assume?

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Fifty Years at a Glance



Adapted from Fig. 1.6 in [1]. We now walk each decade in turn, connecting techniques to what you will implement later in the course.

1970s – Vision as a Sub-Problem of AI

- Vision framed as the perceptual front-end for intelligent robots (MIT, Stanford, CMU).
- Apocryphal but telling: Minsky's 1966 assignment to Sussman to “link a camera to a computer... over a summer” [7] – vision was assumed easy relative to reasoning. It was not.
- Distinguished from classical digital image processing [16] by the goal of recovering **3D structure**, not just 2D signal manipulation.
- **Blocks world & line labeling**: inferring 3D structure from the topology of extracted 2D edges [12, 13, 14, 15].

1970s – Parts, Shading, and Early Correspondence

- **Generalized cylinders & pictorial structures:** solids of revolution / swept curves arranged into part relationships [17] – an early precursor to today's part-based and compositional object models.
- **Intrinsic images:** decomposing an image into surface orientation, shading, and reflectance components [18] – a problem that periodically resurfaces (e.g., intrinsic decomposition with deep networks).
- **Early stereo & optical flow:** feature-based stereo correspondence [19] and intensity-based flow [20, 21] – direct ancestors of Chapters 9 and 12.
- **Structure from motion** begins as a joint recovery problem (Ullman 1979; Longuet-Higgins 1981) – foundational for Chapter 11.

Marr's *Vision* (1982): Consolidating the Decade

- Synthesizes the decade's work under the three-level framework introduced earlier in this session [2].
- Popularizes the “ $2\frac{1}{2}$ -D sketch” – a viewer-centered, intermediate representation between raw image and full 3D model.
- Still cited as a guide for framing new problems today: separate *what* you are computing from *how*.

Callback

Compare this to the Bayesian risk formulation from earlier: Marr's “computational theory” level is precisely where you specify the loss and the constraints before committing to an algorithm.

1980s – Multi-Scale Representations

- **Image pyramids** for blending and coarse-to-fine search [22]; continuous scale-space formulations follow (Witkin 1983).
- **Wavelets** begin displacing/augmenting pyramids in the late 1980s.
- **Shape-from-X family** matures: shading, photometric stereo, texture, focus – multiple cues, one underlying inverse problem (recovering surface orientation/depth).
- **Edge/contour detection** advances (Canny 1986); dynamic contour trackers – **snakes** – introduced [24], plus 3D physically-based deformable models [23].

1980s – Regularization and Probabilistic Models

- Stereo, flow, shape-from-X, and edge detection are unified as **variational optimization** problems, made well-posed via **regularization** (Poggio, Torre & Koch 1985; Terzopoulos 1988) – this is Chapter 4 of the text.
- **Markov random fields (MRFs)** proposed as the discrete counterpart, enabling global search via simulated annealing [25].
- **Kalman-filter-based** online variants for updating uncertainty over time.
- **3D range data** processing (acquisition, merging, recognition) becomes an active subfield in its own right.

Thread to pull

Regularization (deterministic energy) and MRFs (probabilistic energy) are two views of the *same* optimization problem – this equivalence recurs constantly (e.g., diffusion models as iterative denoising / energy descent).

1990s – Structure from Motion Comes of Age

- **Projective reconstruction** without calibration knowledge (Faugeras 1992; Hartley 1994).
- **Factorization methods** for orthographic cameras [26], later extended to the perspective case.
- Convergence with photogrammetry's **bundle adjustment** as the general nonlinear least-squares formulation (Triggs et al. 1999) – this is Chapter 11's core algorithm.
- Fully automated 3D modeling pipelines emerge, culminating later in internet-scale reconstruction [10].

1990s – Correspondence, Tracking, Segmentation, Recognition

- **Optical flow & dense stereo** mature; the biggest breakthrough is global optimization via **graph cuts** [27] – direct predecessor of modern energy-based stereo and segmentation.
- **Multi-view stereo** for complete 3D surfaces; silhouette- and contour-based volumetric reconstruction.
- **Tracking** advances: active contours, particle filters, level sets; face and full-body tracking.
- **Image segmentation** via minimum energy / MDL, normalized cuts [29], mean shift [30].
- **Statistical learning enters vision**: eigenfaces via PCA for face recognition [28] – an early data-driven method, long before deep learning.

1990s – Vision Meets Graphics

- **Image-based modeling and rendering** emerges as a cross-disciplinary field (Chapter 14).
- **Image morphing** (Beier & Neely 1992) → **view interpolation** → **panoramic stitching** [31] → **light fields / Lumigraphs** [48].
- **Image-based modeling**: automatically building realistic 3D models from photo collections.

Discussion prompt

Neural rendering (NeRF and successors, Ch. 14) is often pitched as “new.” What 1990s ideas does it directly continue?

- Image-based rendering techniques (stitching, light fields, HDR) re-branded as **computational photography** as they reach consumer cameras [32].
- **Tone mapping** algorithms make HDR images displayable.
- **Texture synthesis / quilting** [33] and **inpainting** recombine image samples to synthesize new photographs – precursors to today's generative image editing.

- **Feature-based object recognition** combined with learning [34]; pictorial structures revisited [35].
- Interest-point (patch-based) features dominate; contour- and region-based recognition remain active alternatives.
- **Efficient global optimization** matures further: graph cuts, loopy belief propagation for large discrete labeling problems.
- Machine learning applied broadly to vision, enabled by growing web-scale, partially labeled data and increasing compute.

2010s – The Deep Learning Revolution

- **Large labeled datasets:** ImageNet [36, 37], Microsoft COCO [38], LVIS – reliable benchmarks *and* enough data to learn full pipelines end-to-end.
- **GPGPU compute** makes training deep networks practical.
- **AlexNet / SuperVision** [39] wins the 2012 ImageNet challenge – the inflection point after which deep convolutional networks dominate recognition and segmentation [40].
- Rapid extension to video understanding, pose estimation, end-to-end optical flow, denoising, and monocular depth.

2010s – Sensors, SLAM, and the Smartphone

- **Depth sensors:** Kinect (2010) enables real-time 3D modeling [41] and person tracking [43].
- **SLAM and VIO** reach real-time reliability, enabling autonomous flight and mobile AR [42].
- **Computational photography goes mainstream:** panoramic stitching and HDR are now default smartphone camera features; multi-frame low-light denoising and synthetic depth-of-field follow.

Bridge to today

Extrapolate the timeline yourself: what would you put on the 2020s segment? (Transformers in vision, diffusion models, NeRF/3D Gaussian splatting, foundation/vision-language models, self-supervised pretraining.) We will revisit this map throughout the course.

Recurring Historical Themes

- **Representation shift:** hand-designed geometric/photometric models → hand-designed features → learned representations.
- **Optimization shift:** continuous regularization $\leftrightarrow$ discrete MRF/graph-cut inference → gradient-based end-to-end learning.
- **Data shift:** small curated datasets → web-scale labeled datasets → self-/weakly-supervised pretraining.
- **Compute shift:** workstation CPUs → GPGPU → specialized accelerators – hardware constraints (Marr's third level!) repeatedly determine which algorithms become dominant.

Discussion prompt

Which of these four shifts is most likely to define the *next* five years, and what evidence from your own reading supports that?

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Notation Used Throughout the Book

- $\mathbf{v}$ — a **vector**: lower-case bold, always a **column**, so the matrix goes on the left: $\mathbf{M}\mathbf{v}$.
- $\mathbf{M}$ — a **matrix**: upper-case bold. Common ones: $\mathbf{R}$ (rotation), $\mathbf{K}$ (camera), $\mathbf{I}$ (identity).
- T, s — a **single number**: italic.
- $\mathbf{x} = (x, y)$ and $\mathbf{x} = [x \ y]^\top$ mean the same thing.

Homogeneous coordinates (Ch. 2): add one number at the end, marked with a tilde. To go back, divide by it:

$$\tilde{\mathbf{x}} = (6, 4, 2) \longrightarrow \left(\frac{6}{2}, \frac{4}{2}\right) = (3, 2).$$

So $(6, 4, 2)$ and $(3, 2, 1)$ are the **same point** — only the ratios matter.

Cross product as a matrix: $\mathbf{a} \times \mathbf{b} = [\mathbf{a}]_{\times} \mathbf{b}$, which lets us treat it like any other matrix multiplication.

Watch out

Other books use different symbols for all of this [44]. Leal be easy to follow.

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Key Takeaways

- Computer vision = inverting an ill-posed forward (image formation) model; every method injects some form of prior/regularizer.
- Four methodologies – scientific, statistical, engineering, data-driven – are complementary lenses, not competing schools.
- Marr's three levels (computational theory / representation & algorithm / hardware) remain a useful way to structure both papers and your own research.
- The field's 50-year history shows recurring shifts in representation, optimization, data, and compute – deep learning is the latest, not the last, such shift.
- The textbook's chapters are loosely coupled along a 2D/3D axis; plan your reading path accordingly.

Open Questions for Discussion

- ❶ Where does your own research problem sit on the “what vs. where” axis, and which chapters are you implicitly assuming?
- ❷ Which of the four methodological pillars is currently under-used in your subfield, and would combining it with another help?
- ❸ Pick one 1980s–1990s technique (regularization, MRFs, graph cuts, bundle adjustment) and identify its direct descendant in a 2020s paper you have read.
- ❹ Marr’s “hardware implementation” level shaped which algorithms won historically (GPUs → deep learning). What upcoming hardware shift (e.g., edge accelerators, neuromorphic chips) might shift the field again?

Course code, exercises, and materials:
https://github.com/CarlosMenacho/computer_vision_class



Chapter 2: Image Formation

Geometric primitives & transformations · Photometric image formation · The digital camera

Before next class: read Szeliski Ch. 2.1–2.2 and come with one question connecting today's “forward model” discussion to a specific camera projection equation.

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